Band-1 Cap-set a(7) Lower-Bound Witness Re-cert

Finite lower-bound witness / verification only (exact a(7) OPEN; NOT asymptotic cap-set solved; GATE-04 LOCKED)

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parent lab. `/workspace/experiments/band1/capset_exact/` (a(0..3) exact; $a(4) \geq 20$; lit $a(7) \geq 236$).
Lab. `/workspace/experiments/band1/capset_a7_lower_witness/`.
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite lower-bound witness / verification** increment only. It **does not** claim the asymptotic cap-set problem solved (Ellenberg–Gijswijt is asymptotics). It **does not** claim exact $a(7)$ known or closed — **exact** $a(7)$ **remains OPEN**. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. `clay_closed = false`.

Abstract

We re-certify explicit capset lower-bound witnesses in $(\mathbb{Z}/3\mathbb{Z})^n$. Box enumeration recovers exact $a(0), \ldots, a(3) = 1, 2, 4, 9$. Explicit verified point lists give $a(4) \geq 20$, $a(5) \geq 45$, $a(6) \geq 112$. The **primary target** is $a(7)$: extracting the largest affine subcap from Yves Edel's published 248-cap in $\mathrm{PG}(7, 3)$ yields a verified 236-point capset in $\mathrm{AG}(7, 3)$, matching the literature floor (OEIS A090245 / MinT). **Exact $a(7)$ remains OPEN** (literature interval $236 \leq a(7) \leq 291$).

Keywords. Cap set; $\mathrm{AG}(n, 3)$; Hill cap; Edel–Bierbrauer; lower-bound witness; Band-1; non-claim.

1. Statement lock

Definition. A set $C \subset (\mathbb{Z}/3\mathbb{Z})^n$ is a **capset** iff it contains no three distinct points x, y, z with $x + y + z \equiv 0 \pmod 3$ coordinatewise (equivalently: no 3-term arithmetic progression / affine line).

Notation. $a(n) = \max |C|$ over capsets $C \subset (\mathbb{Z}/3\mathbb{Z})^n$.

Finite lemma (this increment only). Explicit on-box certificates:

$$a(0..3) \text{ exact}; \quad a(4) \geq 20;\ a(5) \geq 45;\ a(6) \geq 112;\ a(7) \geq 236.$$

Not asymptotic resolution; **not** exact $a(7)$.

2. Methods

2.1 Exact $a(0..3)$

Branch-and-bound over bitset candidates with 0 fixed (translation invariance of maximum size). Recovers literature values $1, 2, 4, 9$.

2.2 $a(4) \geq 20$

Explicit 20-point witness (Pellegrino size; reused from prior `capset_exact` lab). Capset predicate verified pairwise via third-point $c = -(x + y)$.

2.3 $a(5) \geq 45$ from Hill 56-cap

Bierbrauer's explicit Hill cap in $\mathrm{PG}(5, 3)$: weight-6 points with even number of 2's (16 projective points) union weight-3 points on ten specified triads (40 points). Scan all nonzero linear forms; maximal affine chart has size 45. Verified as an $\mathrm{AG}(5, 3)$ capset. Citations: Hill 1973; Edel–Ferret–Landjev–Storme 2002; MinT Hill-cap note.

2.4 $a(6) \geq 112$ lift

From the 45-cap T: place T in hyperplane $x_6 = 0$, $-T$ in $x_6 = 1$, and a greedy compatible 22-set M in $x_6 = 2$. Yields a verified 112-point capset (Potechin size; $\{45, 22, 45\}$-type profile as in Thackeray

arXiv:2206.09804).

2.5 Primary: $a(7) \geq 236$ from Edel 248-cap

Fetch / embed Yves Edel's published generator matrix for a 248-cap in PG$(7,3)$ (`(248,7,3).html`). Columns are homogeneous coordinates. For linear form $L = e_8$ (Edel's maximal-weight last row), the 236 points with $L \cdot p \neq 0$, dehomogenized, form an AG$(7,3)$ capset. Exhaustive scan over all $3^8 - 1$ forms confirms max affine size $= 236$ (MinT: largest affine sub-cap 236).

Weaker constructive alternative (not needed): product/doubling $2 \cdot a(6) = 224$ leaves a gap of 12 to the lit floor; this run **closes that gap on-box**.

3. Results

n	Box result	Literature	Status
0..3	exact $1, 2, 4, 9$	$1, 2, 4, 9$	**PASS exact**
4	≥ 20	$= 20$	**PASS lower**
5	≥ 45	$= 45$	**PASS lower**
6	≥ 112	$= 112$	**PASS lower**
7	≥ 236	≥ 236, exact OPEN	**PASS lower; exact OPEN**

Wall clock (box): ≈ 1.09s for full `verify_capset_a7.py` (ALL PASS).

4. What is still OPEN

- **Exact** $a(7)$ (and all $n \geq 7$).
- Asymptotic cap-set density (Ellenberg–Gijswijt upper bound is asymptotic, not a small-n exact value).
- Literature upper $a(7) \leq 291$ is **not** re-proven here — recorded as ledger only.

5. Reproducibility

```
cd /workspace/experiments/band1/capset_a7_lower_witness
python3 verify_capset_a7.py
# writes results.json, literature_ledger.json, SUMMARY.txt
```

Data: `data/edel_248_PG73.html` (Edel matrix snapshot); witness code lists under `data/a*_codes.txt`.

6. Honesty coda

No Clay-closed stamp. No finished-ToE certificate. GATE-04 collection **LOCKED**. No invented J. Label: **finite lower-bound witness / verification only**.